

Reminder: Empirical and simulation exercises must be submitted with “stand-alone” code and data (if any) in an appropriate format. The homework is due Sunday (data, code, and word-processed report by email to **the instructor before 10 PM Sunday**). Also, submit your work-processed report via **Gradescope**.

Consider a four-variable VAR using quarterly observations: $\Delta\log(\text{real GDP})$, $\Delta\log(\text{GDP deflator})$, $\Delta\log(\text{EXR})$, and FFR, where EXR denotes a trade-weighted dollar exchange rate. To minimize the effects of the binding zero lower bound, you may use the sample up to 2008:q4.

1. Select the lag length using BIC, AIC, and likelihood ratio tests. Compare the differences.
2. Given the optimal lag length selected by AIC, estimate the model by least squares.
3. Using Cholesky decomposition and the ordering [$\Delta\log(\text{real GDP})$, $\Delta\log(\text{GDP deflator})$, $\Delta\log(\text{EXR})$, FFR],
 - i. Compute and plot impulse responses of $\log(\text{real GDP})$, $\log(\text{GDP deflator})$, $\log(\text{EXR})$, and FFR (horizon $h = 0, \dots, 20$) to a one-standard-deviation shock in FFR. (HINT: You can obtain the response of $\log(\text{real GDP})$ by cumulating the responses of $\Delta\log(\text{real GDP})$).
 - ii. For each impulse response, construct, plot, and compare error bands (use 5th and 95th percentiles) using:
 - Parametric Monte-Carlo (HINT: You may want to use the half-vectorization routine, `vech.m`, and the duplication matrix, `dupmat.m`, uploaded on Canvas),
 - Bootstrap.
 - iii. Does the price level increase or decrease in response to the contractionary monetary policy shock (i.e. after an increase in FFR)? How can you reconcile an increase (decrease) in the price level? You may want to consult Sims (EER, 1992).
 - iv. Compute and plot the contribution of the FFR shocks to variation in $\log(\text{real GDP})$, $\log(\text{GDP deflator})$, $\log(\text{EXR})$, and FFR (the forecast variance decompositions) at horizons $h = 0, \dots, 20$.
4. Repeat 3.i using a different ordering [$\Delta\log(\text{EXR})$, $\Delta\log(\text{real GDP})$, $\Delta\log(\text{GDP deflator})$, FFR]. Compare your results with that in 3.i. You may want to consult Christiano, Eichenbaum, and Evans (Handbook of Macroeconomics, 1999).
5. Compare your empirical responses with the predictions of the Dornbusch (“overshooting”) model. You may want to consult Eichenbaum and Evans (QJE, 1995).